

Western New England University
College of Engineering
ECE Department
EE Lab I
EE 319
Fall 2024
Lab #01 – HPF Measurements

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	Score	Max
Circuit Set-Up		100
Lab Practices		100
Magnitude Bode Plots		100
Phase Bode Plots		100
Tables		100
Deductions		----
Total		500

High-Pass Filter Tasks

1. Construct the high-pass filter as shown in Figure HPF-1. Assume that $R_1 = 10\text{ k}\Omega$, $R_2 = 10\text{ k}\Omega$, and $C_2 = 56\text{ nF}$.

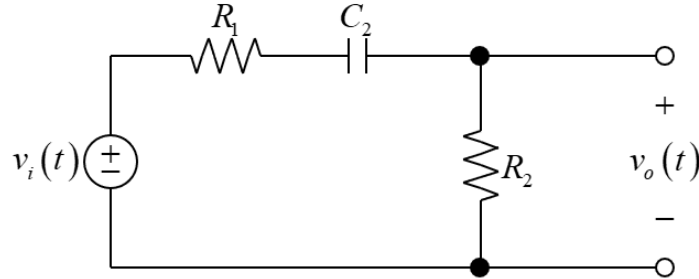
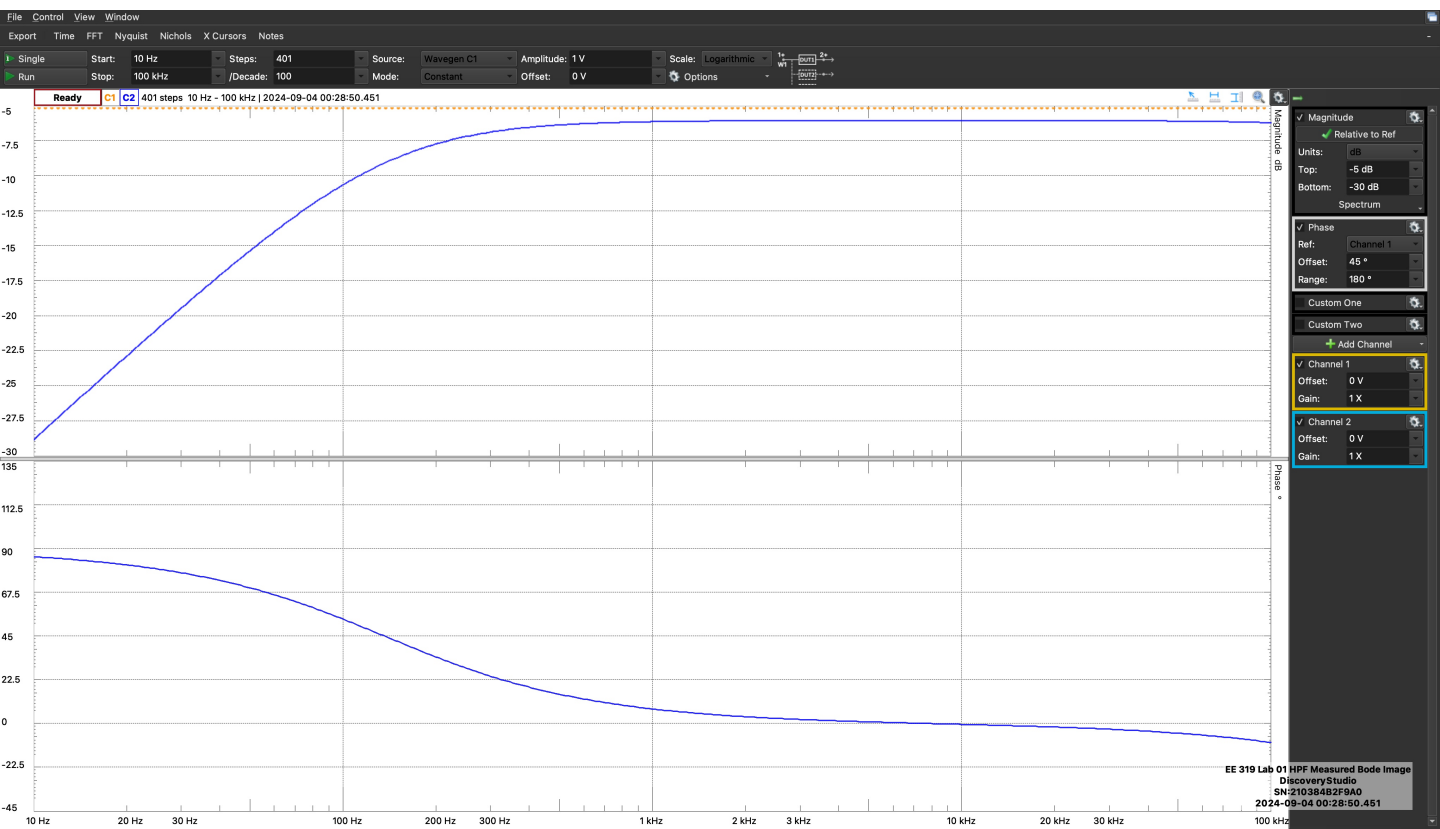
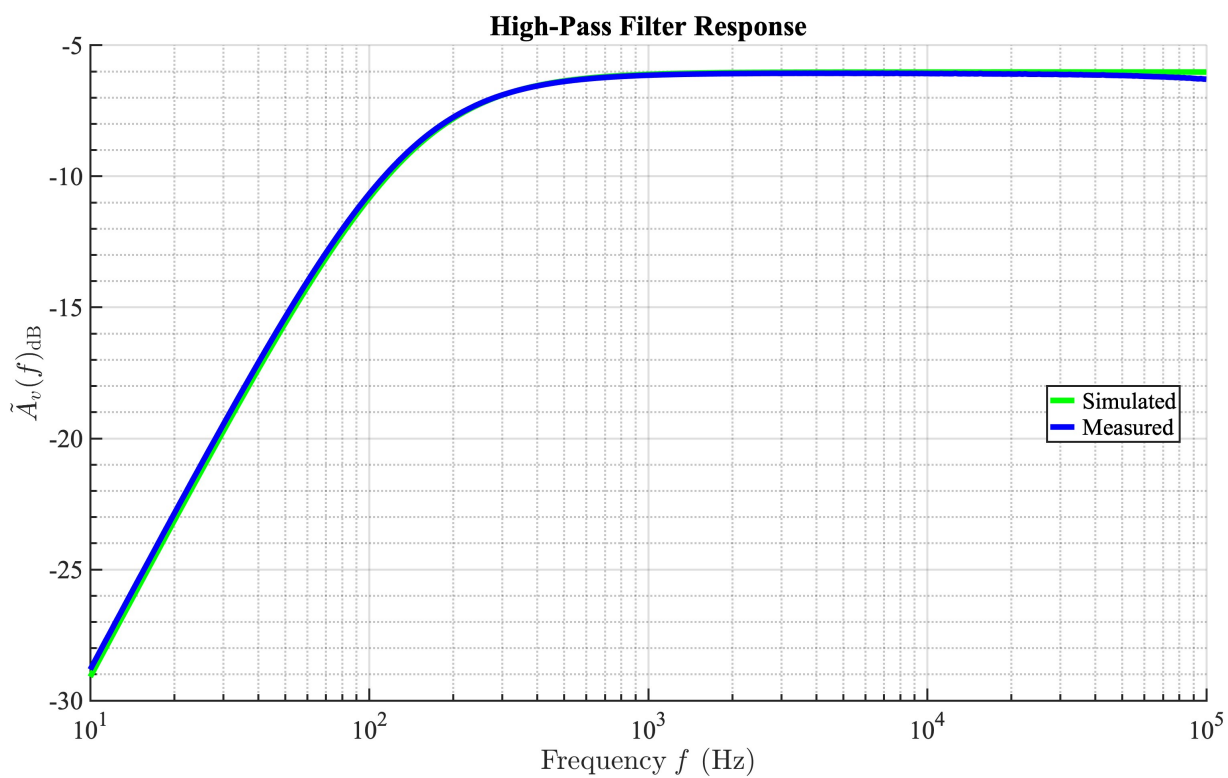


Figure HPF-1. Schematic of a high-pass filter.

2. Measure the Bode plot for the gain $\tilde{A}_v(f) = \tilde{V}_o(f)/\tilde{V}_i(f)$. Use a frequency range of 10 Hz to 100 kHz.
3. Using MATLAB, plot the magnitude of gain, $|\tilde{A}_v(f)|$, in dB, for the measured response of the high-pass filter. Employ a range of 10 Hz to 100 kHz for the frequency and a range of -30 to -5 dB for the gain.
4. Using MATLAB, plot the phase of the gain, $\theta(f)$ in degrees, for the measured response of the high-pass filter. Employ a range of 10 Hz to 100 kHz for the frequency and a range of 0° to $+90^\circ$ for the phase.
5. Determine a value for the high-frequency gain, $A_{v(HF)}$; and the 3 dB frequency, f_L . Complete the measured column in Table HPF-1.
6. At a frequency of $f = f_L$, measure $v_i(t)$ and $v_o(t)$ versus t .
7. At a frequency of $f = 100\text{ kHz}$, measure $v_i(t)$ and $v_o(t)$ versus t .
8. At a frequency of $f = 10\text{ Hz}$, measure $v_i(t)$ and $v_o(t)$ versus t .

EE 319 — Lab 01 — High-Pass Filter — Measured Bode Plot





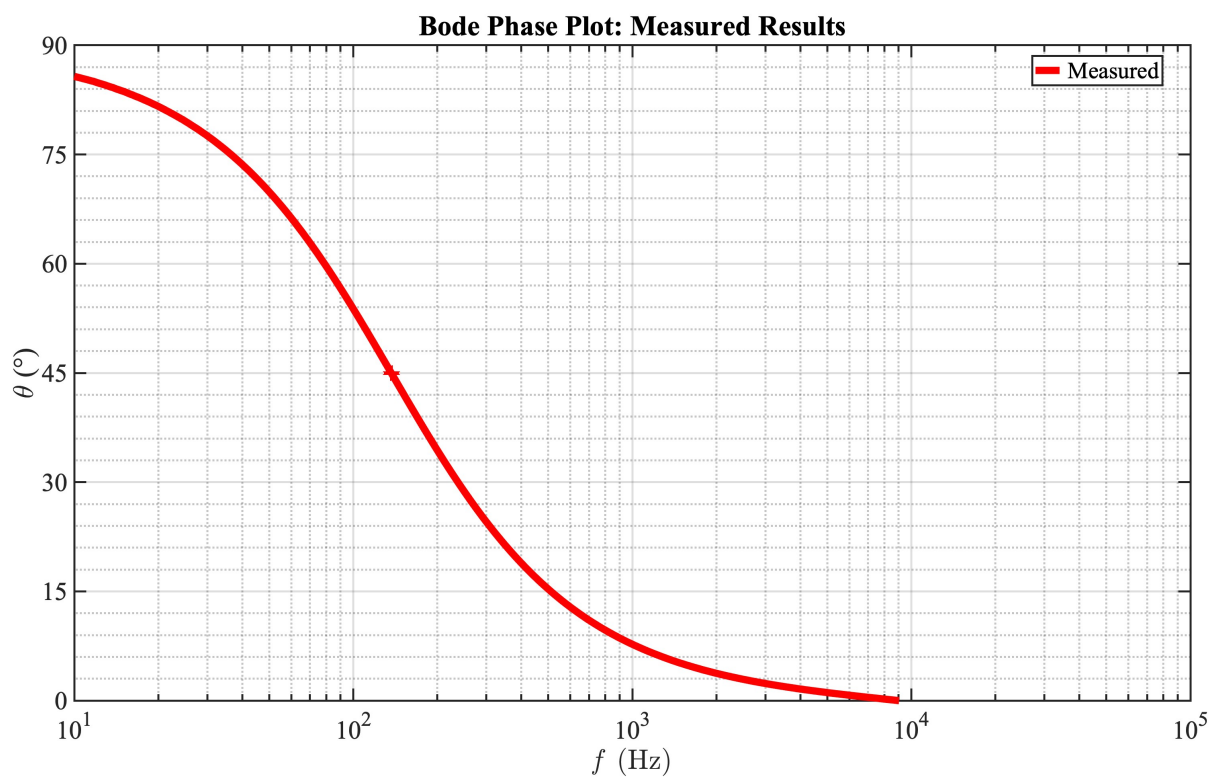


Table HPF-1 Calculated, simulated, and measured parameters of the high-pass filter.

	Calculated	Simulated	Measured	
$A_{v(HF)}$	500.0000	499.9995	496.4600	mV/V
$A_{v(HF)}$	-6.0206	-6.0206	-6.0823	dB
f_L	142.1026	142.1023	136.8137	Hz

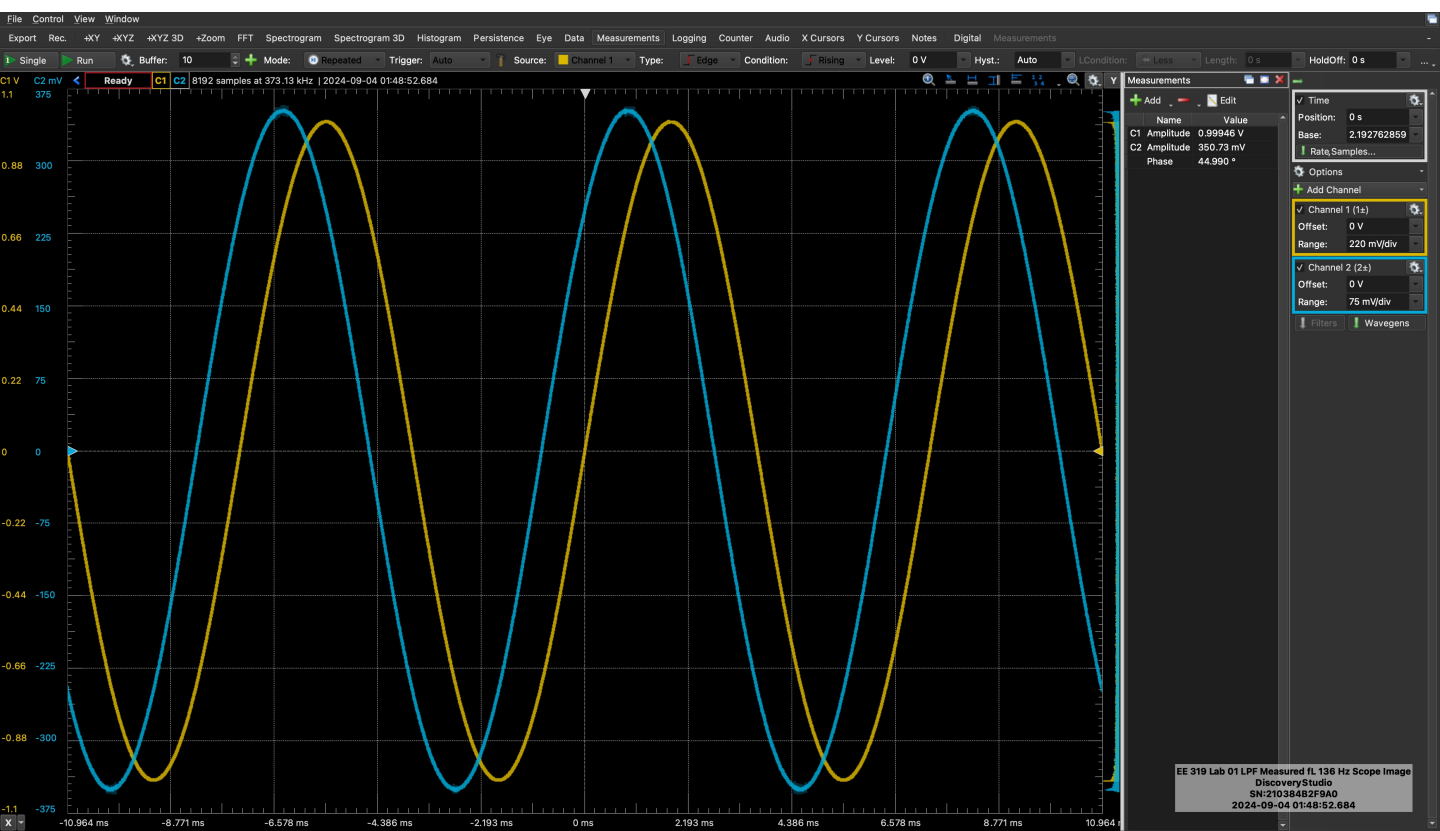
EE 319 — Lab 01 — High-Pass Filter — Comparison of Calculated & Simulated Values

	Calculated	Simulated	Units	Calc Sim Diff	Calc Sim % Diff
Av(LF)	500.0000	499.9995	mV/V	0.0005	0.0001
Av(LF) dB	-6.0206	-6.0206	dB	0.0000	0.0000
fH	142.1026	142.1023	kHz	0.0003	0.0002

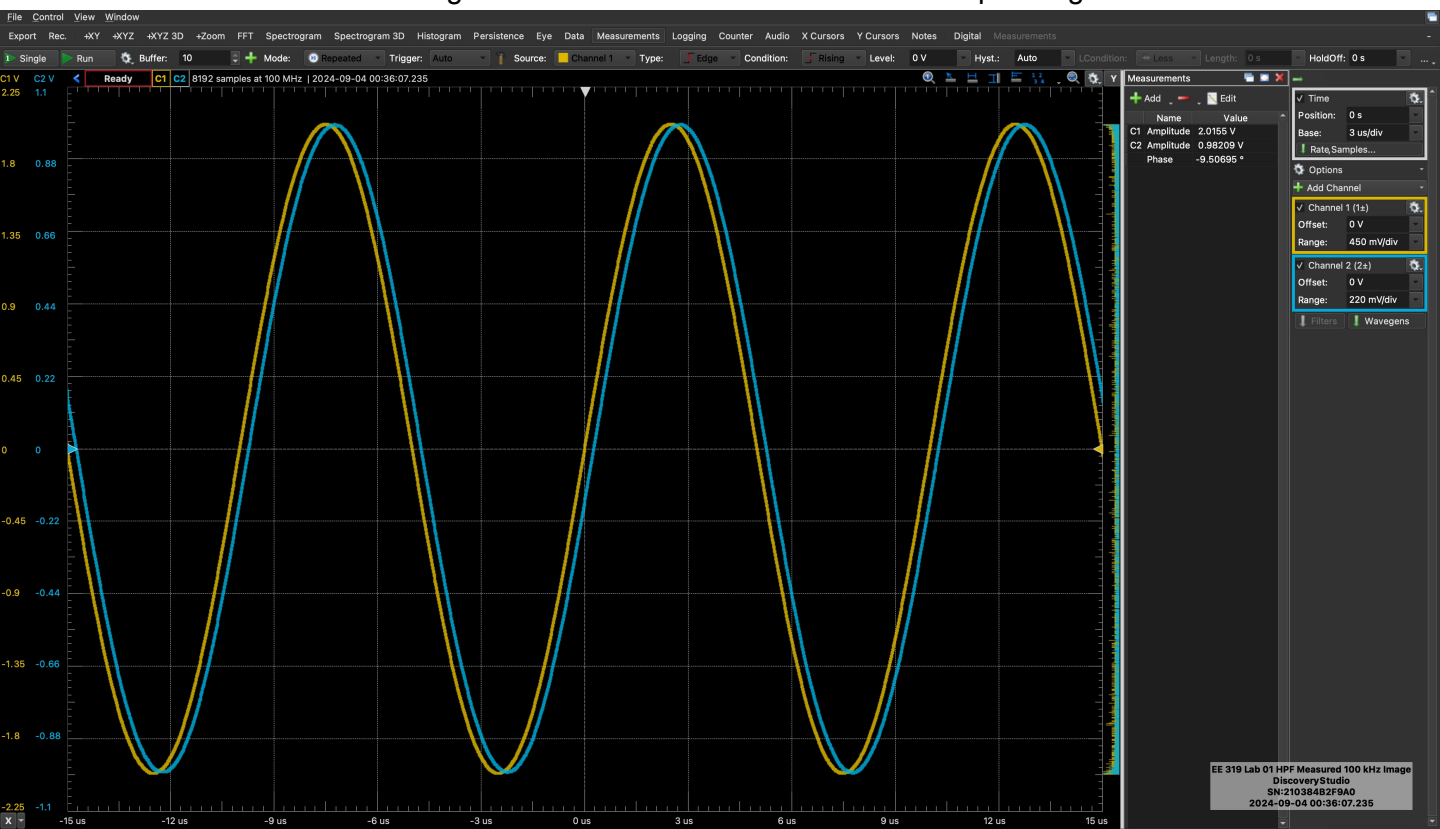
EE 319 — Lab 01 — High-Pass Filter — Comparison of Calculated & Measured Values

	Calculated	Measured	Units	Calc Meas Diff	Calc Meas % Diff
Av(LF)	500.0000	496.4600	mV/V	3.5400	0.7080
Av(LF) dB	-6.0206	-6.0823	dB	0.0617	1.0248
fH	142.1026	136.8137	kHz	5.2889	3.7219

EE 319 — Lab 01 — High-Pass Filter — Measured $f_L = 136.8187$ Hz Scope Image



EE 319 — Lab 01 — High-Pass Filter — Measured 100 kHz Scope Image



EE 319 — Lab 01 — High-Pass Filter — Measured 10 Hz Scope Image

